Practice workbook · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Defining and using gene, allele, dominant, recessive, homozygous, heterozygous, genotype, phenotype and codominance; listing the genotypes for a blood group; and explaining how carriers pass on a recessive condition.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-GENETIC-TERMS

1. Question 7 · part a.i · 2 marks

June 2023 Paper 1R

Worked-example retry · Attempt part a.i only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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1. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 9 · part a.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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2. Question 9 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 8 · part a.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.i only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

.....

(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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3. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 8 · part a.ii · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.ii only.

8 (a) Genetic conditions are often caused by recessive alleles.

(i) Which of these describes an allele?

(1)

- ☒ **A** a chromosome that controls a characteristic
- ☒ **B** an alternative form of a gene
- ☒ **C** a pair of chromosomes that are the same size
- ☒ **D** one of a pair of chromosomes

(ii) State what is meant by the term **recessive**.

(1)

.....

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(b) A student investigates a genetic condition found in his family.

The student made these observations.

- he is male and has the condition
- his mother does not have the condition
- his father and his father's brother have the condition
- his father's sister does not have the condition
- his father's mother (grandmother) does not have the condition but his father's father (grandfather) does have the condition

(i) Construct a family pedigree showing which members of the family have the condition and which do not.

(3)

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P 6 7 0 6 2 R A 0 2 6 2 8

4. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a · 2 marks

November 2023 Paper 2

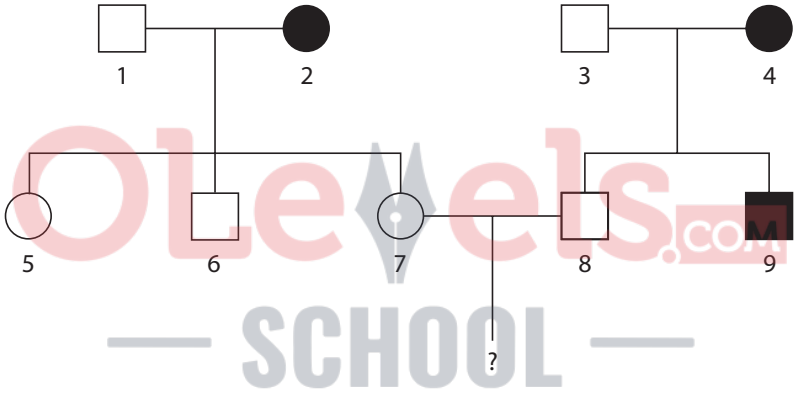
Further practice · Attempt part a only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



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Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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5. Question 5 · part a · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

.....

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

.....

(iii) Give one symptom of polydactyly.

(1)

.....

.....

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

.....

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6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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7. Question 3 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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7. Question 3 · part b.i · 1 mark

June 2024 Paper 1

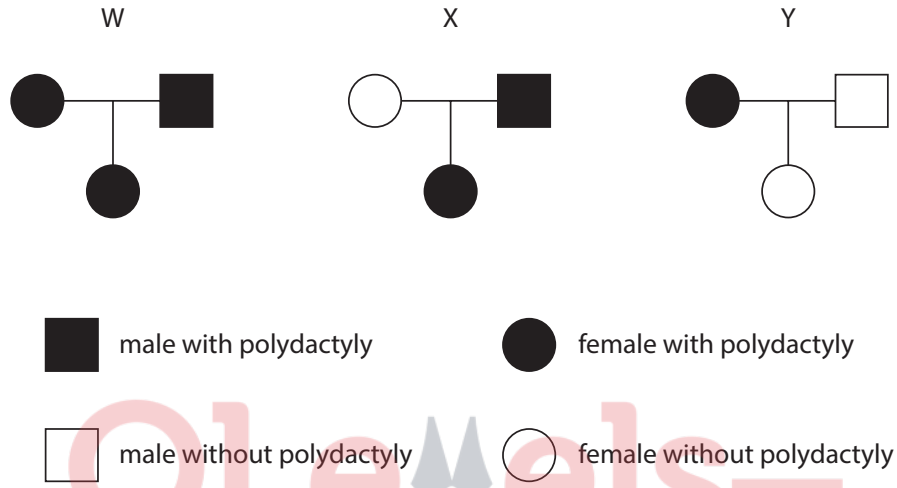
Further practice · Attempt part b.i only.

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(ii) The family pedigree diagrams show the inheritance of polydactyly.



Which of the diagrams, W, X or Y, gives the family pedigree for parents R and S?

(1)

(c) A gene is a short section of DNA.

Describe the structure of DNA.

(4)

(Total for Question 3 = 9 marks)

7. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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8. Question 4 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

8. Question 4 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(ii) Explain why diploid cells contain two alleles for each characteristic.

(4)

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

(2)

(ii) Give one symptom of cystic fibrosis.

(1)

(Total for Question 4 = 11 marks)

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8. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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9. Question 6 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

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9. Question 6 · part a · 1 mark · Working and answer

Working and answer

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10. Question 9 · part a.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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10. Question 9 · part a.ii · 1 mark · Working and answer

Working and answer

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11. Question 7 · part a.ii · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

7 The four blood groups that occur in humans are group A, group B, group AB and group O.

There are three blood group alleles: I^A, I^B and I^O.

I^O is recessive to both I^A and I^B.

I^A and I^B are co-dominant to each other.

(a) (i) Explain what is meant by the term **co-dominant**.

(2)

(ii) State the possible genotypes of a person with blood group B.

(2)

(b) Haemophilia is a sex-linked condition that prevents the blood clotting.

Explain why males are more likely to be affected by haemophilia than females.

(4)

(Total for Question 7 = 8 marks)

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11. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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12. Question 8 · part a.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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12. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

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13. Question 8 · part b.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.i only.

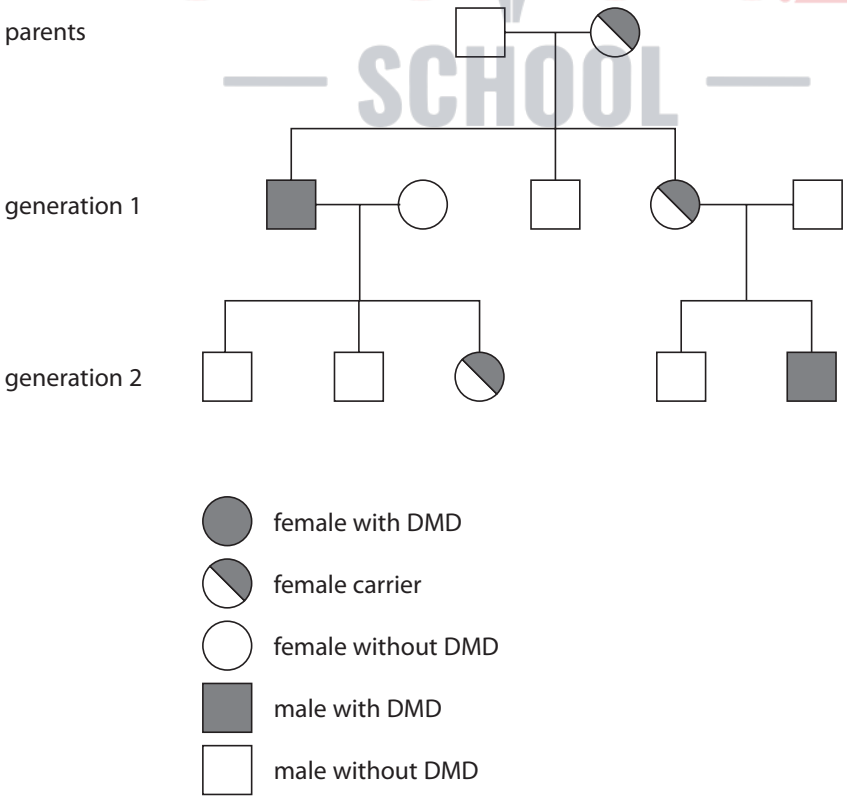
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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13. Question 8 · part b.i · 2 marks

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Further practice · Attempt part b.i only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

13. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

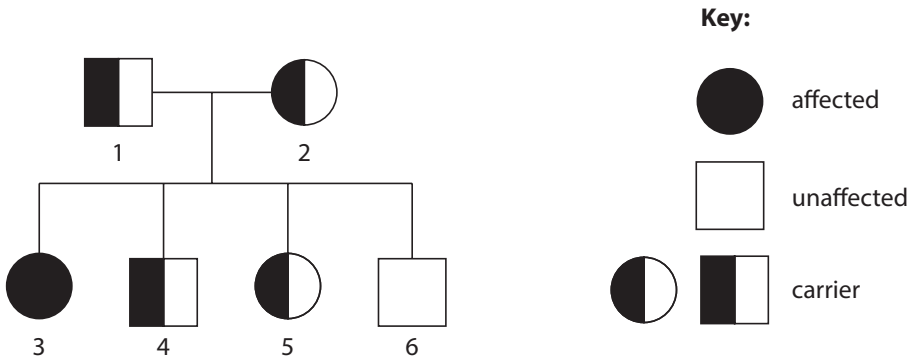
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14. Question 8 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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14. Question 8 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

(Total for Question 8 = 11 marks)

14. Question 8 · part a.iii · 2 marks · Working and answer

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15. Question 6 · part b.i · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

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15. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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16. Question 6 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(1)

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Describe what causes an individual to have cystic fibrosis.

(2)

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(4)

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16. Question 6 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(iii) Explain why two parents who do not have cystic fibrosis can have a child who has cystic fibrosis.

(2)

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(Total for Question 6 = 9 marks)

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Working and answer